Project Executable Files

Maximum Marks: 2 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 | ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

How to Run the Rising Waters Application

Follow the step-by-step instructions below to set up the environment and run the Rising Waters flood prediction web application locally or on IBM Cloud.

Step	Action	Command / Details
1	Clone the Repository	git clone https://github.com/Godesivaramakrishna/Flood-Management-System-by-skill-wallet.git
2	Create Virtual Environment	conda create -n flood-env python=3.11 OR python -m venv .venv
3	Activate Environment	conda activate flood-env OR .venv\Scripts\activate (Windows)
4	Install Dependencies	python -m pip install -r requirements.txt
5	Train Model (Optional — pre-trained model included)	python model/generate_dataset.py && python model/train_model.py
6	Run Flask Application	python app.py
7	Open in Browser	http://127.0.0.1:5000
8	Run with Docker	docker build -t rising-waters . && docker run -p 5000:5000 rising-waters

Key Executable Files

File	Description
app.py	Main entry point — run this to start the Flask web server
model/floods.save	Pre-trained XGBoost model (Joblib) — 96.55% test accuracy
model/scaler.save	Pre-fitted StandardScaler (Joblib) — must match floods.save
model/train_model.py	Retrain all 4 models and regenerate floods.save + scaler.save
requirements.txt	All Python dependencies with pinned versions
Dockerfile	Docker container build configuration
manifest.yml	IBM Cloud Foundry deployment manifest
Procfile	Heroku/IBM Cloud process type declaration